HOW TAG-TO-TAG MATCHING WORKS IN IAM

In ACME's confidential document policy, the condition that checks whether the principal's department matches the owner's department is:

```
"StringEquals": {
  "aws:PrincipalTag/department":
     "${s3:ExistingObjectTag/owner-department}"
}
```

Here's how each side of the comparison is populated and evaluated:

The `aws:PrincipalTag/<attribute-name>` tag is a global condition key used by IAM to reference tags attached to the requesting principal.

- In ACME's setup, AWS IAM Identity Center connects to the corporate directory.
- The `department` attribute from the directory is mapped to a principal tag in Identity Center.
- When the user logs in, Identity Center passes this attribute into the AWS session as `aws:PrincipalTag/department`.
- Because principal tags are attached at sign-in, they consistently display the current state of the identity source.

The construct `${s3:ExistingObjectTag/owner-department}` is a policy variable that IAM replaces at runtime with the value of the owner-department tag on the specific S3 object being accessed.

- ACME's upload process automatically adds this tag when a document is stored in S3, assigning it to the uploader's department.
- Variable substitution is needed here because IAM must look up the tag on the specific resource being accessed in the request.

When a request is made, IAM compares the principal's `department` tag to the S3 object's `owner-department` tag. If they match and all other conditions in the statement are met, the action is allowed. Since both tags ultimately come from the same authoritative source (ACME's corporate